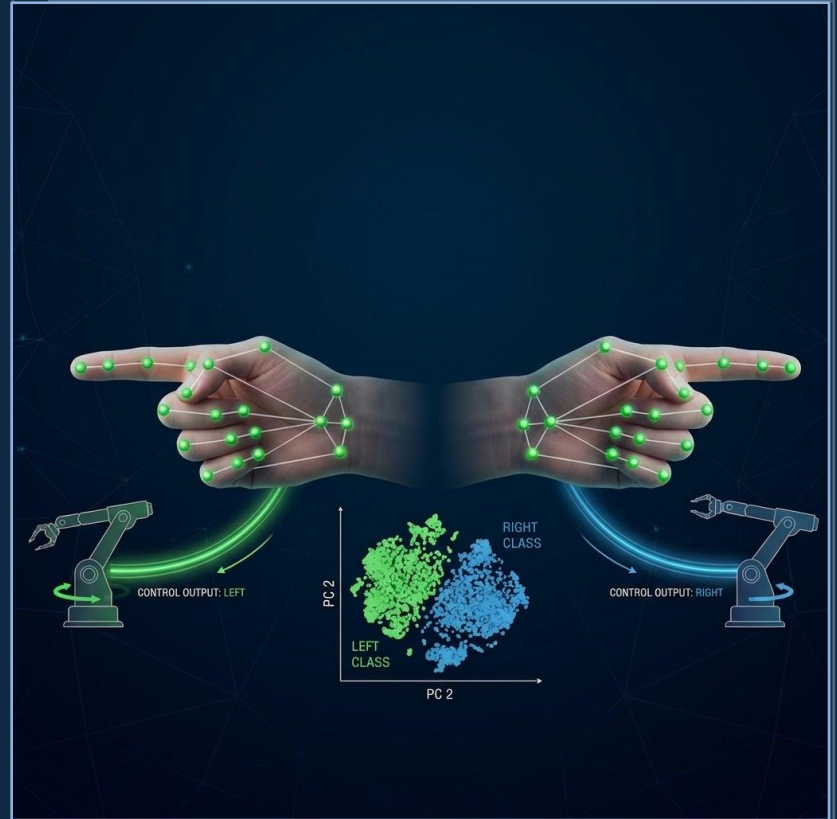


EE 344 Final Project

Pointing Direction Classification from 3D Hand Landmarks

From unsupervised exploration (t-SNE +
K-means) to a supervised classifier pipeline

Team: Torres, Jonathan



Final project presentation

Goal: Classify Hand Pointing Direction (Left vs. Right) using 3D Landmark Coordinates.



Application Domain: Robotics

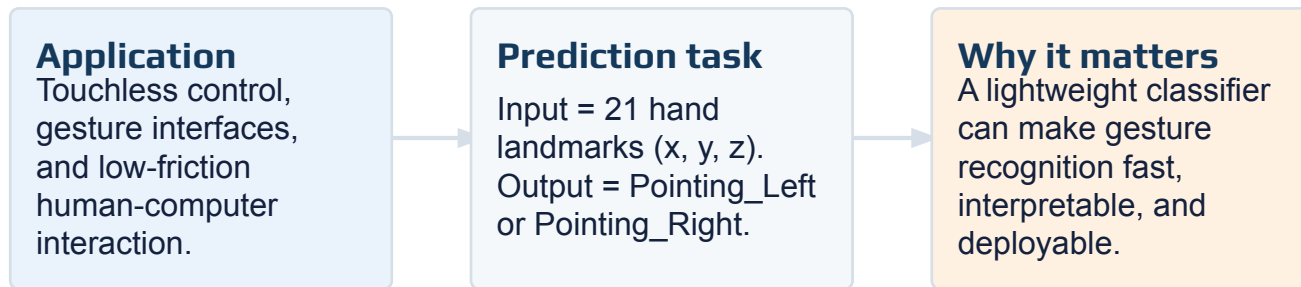
Intuitive, Sensorless Control
for autonomous hardware.



Human-Computer Interaction (HCI)

Transitioning to low-friction,
vision-based gesture
interfaces.

Importance: Reduces hardware complexity, cuts costs, and makes interaction natural.



- This is a binary classification problem built from landmark-based hand pose data.
- We wanted a model that is accurate, simple to explain, and efficient enough for practical use.
- Process: start with unsupervised exploration, then move to supervised learning for stronger predictive performance.

Each record represents one hand pose encoded by the 3D coordinates of 21 landmarks.

603

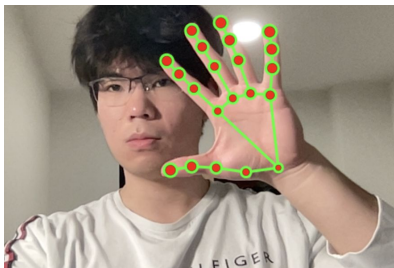
samples

63

numerical features

2

classes



Feature layout

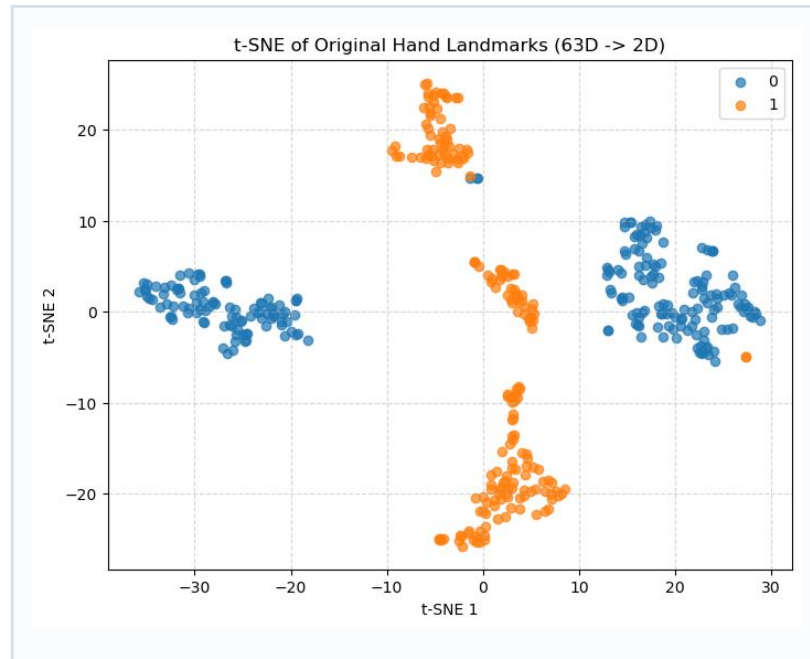
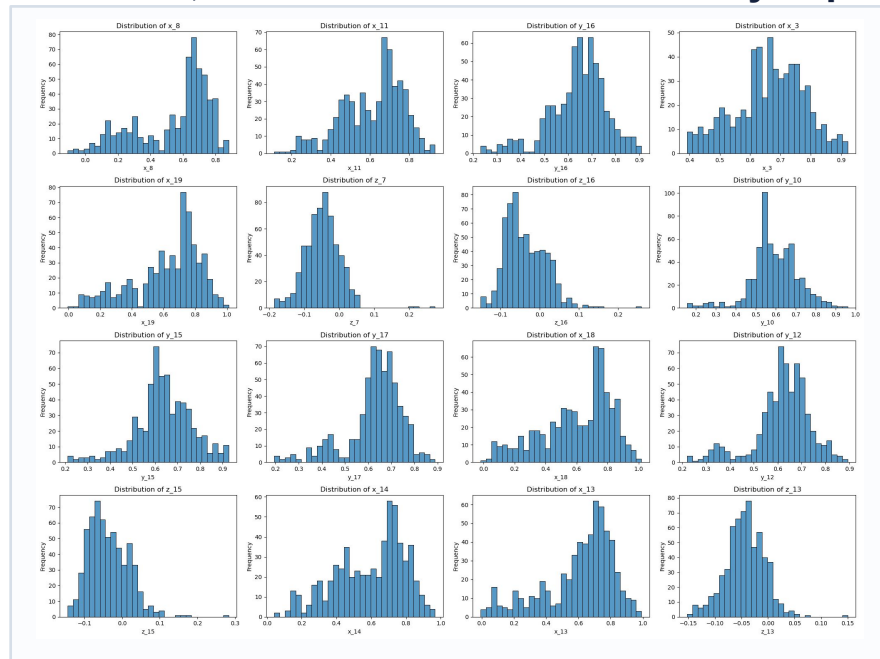
`x_0 y_0 z_0``:``x_20 y_20``z_20``+ target:``direction`

Label classes:

Pointing_Left and
Pointing_Right

- Data shape in the notebook: 603 rows × 64 columns before splitting features and target.
- Target encoding produced two one-hot columns: direction_Pointing_Left and direction_Pointing_Right.
- Train/test split in the notebook: 422 training examples and 181 test examples.

Two observations guided the modeling choices: many features looked roughly Gaussian, and the classes were visually separable.



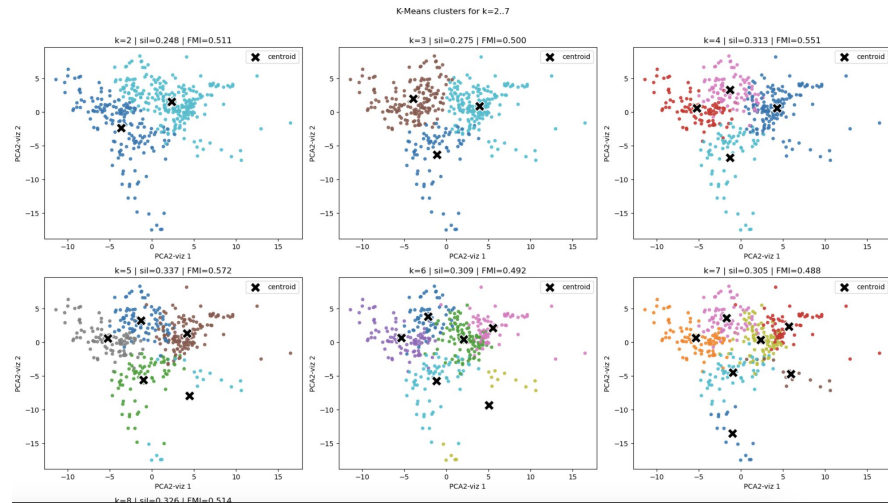
EDA takeaways

- Approximate Gaussian feature shapes supported the use of StandardScaler.
- t-SNE showed meaningful class structure, so the landmarks clearly carried signal.
- The exploratory view encouraged both clustering experiments and later supervised classification.

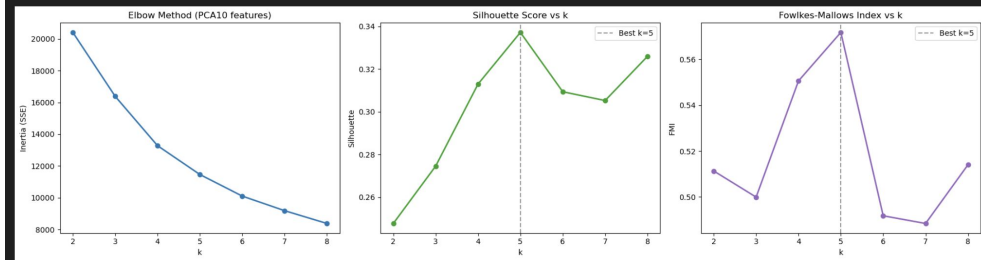
K-mean Clustering Result

1. **Objective:** Determine if the 63-dimensional landmark data naturally separates into two distinct directional groups without using true labels.
2. **Elbow Method Findings:** The inertia plot lacks a sharp "elbow" at $k=2$, instead suggesting optimal clustering around $k=4$ or $k=5$.
3. **Conclusion:** Unsupervised, distance-based clustering is insufficient for isolating the Left/Right classes, strictly necessitating supervised learning pipelines.

	k	inertia	silhouette	fowlkes_mallows
0	2	20404.6435	0.2477	0.5113
1	3	16377.5774	0.2746	0.4999
2	4	13270.6630	0.3129	0.5506
3	5	11455.6984	0.3371	0.5718
4	6	10097.7883	0.3094	0.4918
5	7	9179.1725	0.3052	0.4884
6	8	8374.2773	0.3259	0.5140



$k=8$ | sil=0.326 | FMI=0.514



We kept the pipeline simple and interpretable before comparing models.

Clean

Drop filename
Coerce numeric
Check missing
values

Split

Stratified 70/30
422 train
181 test

Scale

StandardScaler
fit on train
transform test

Encode

One-hot target
Left vs Right

Reduce

PCA to 10 comps
for a lighter
feature set

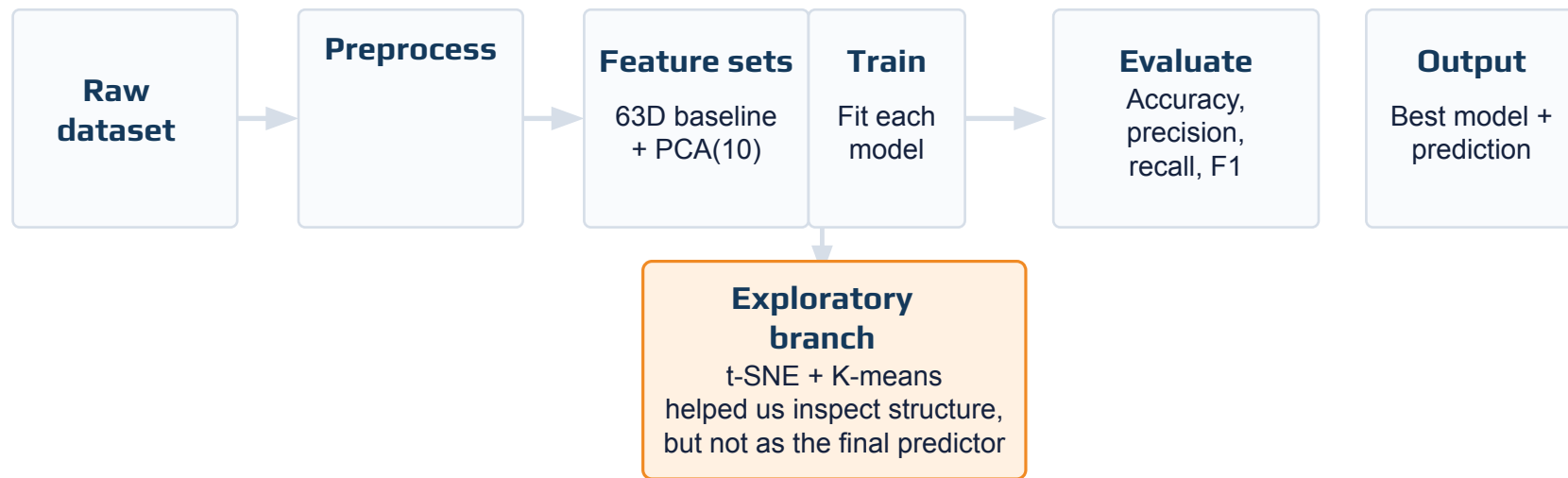
Why scaling mattered

- Coordinate features have different variances and ranges.
- Standardization prevents a few high-variance coordinates from dominating distance- or margin-based models.

Why PCA was tested

- It compressed 63 correlated coordinates into 10 components.
- That made the final comparison about accuracy vs. complexity, not only raw performance.

Our workflow combines exploratory unsupervised analysis with a supervised training-and-selection stage.



Pipeline summary: exploratory unsupervised methods were useful for understanding the geometry of the data, but the final model selection came from supervised train/test evaluation.

We treated model selection as a progression: explore first, then compare supervised classifiers on the same processed data.

Attempt 1 — Unsupervised

- t-SNE visualized the geometry of the 63D landmark space.
- K-means was tried as an early clustering approach.
- Useful for exploration, but not strong enough for final prediction.

Takeaway: strong for structure discovery; not our final classifier.

Attempt 2 — Supervised classifiers

- Logistic Regression and Random Forest were the main contenders.
- SVM, KNN, Perceptron, Decision Tree, etc.

Selection logic

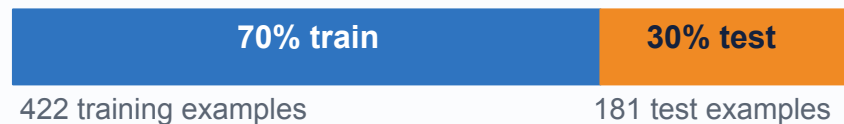
- Random Forest had the highest test F1 on the full 63-feature baseline.
- After PCA reduction, Logistic Regression became the top model while keeping the pipeline simpler.
- So Random Forest was a strong benchmark, but Logistic Regression became our preferred final presentation model.

We compared models under the same split and the same metrics, then checked whether PCA changed the ranking.

Evaluation setup

- Train/test split: 70% / 30%, stratified.
- Models were fit on scaled training data and evaluated on the held-out test set.
- We reported accuracy, precision, recall, and F1 score.
- We repeated the same comparison on the PCA-reduced feature set.

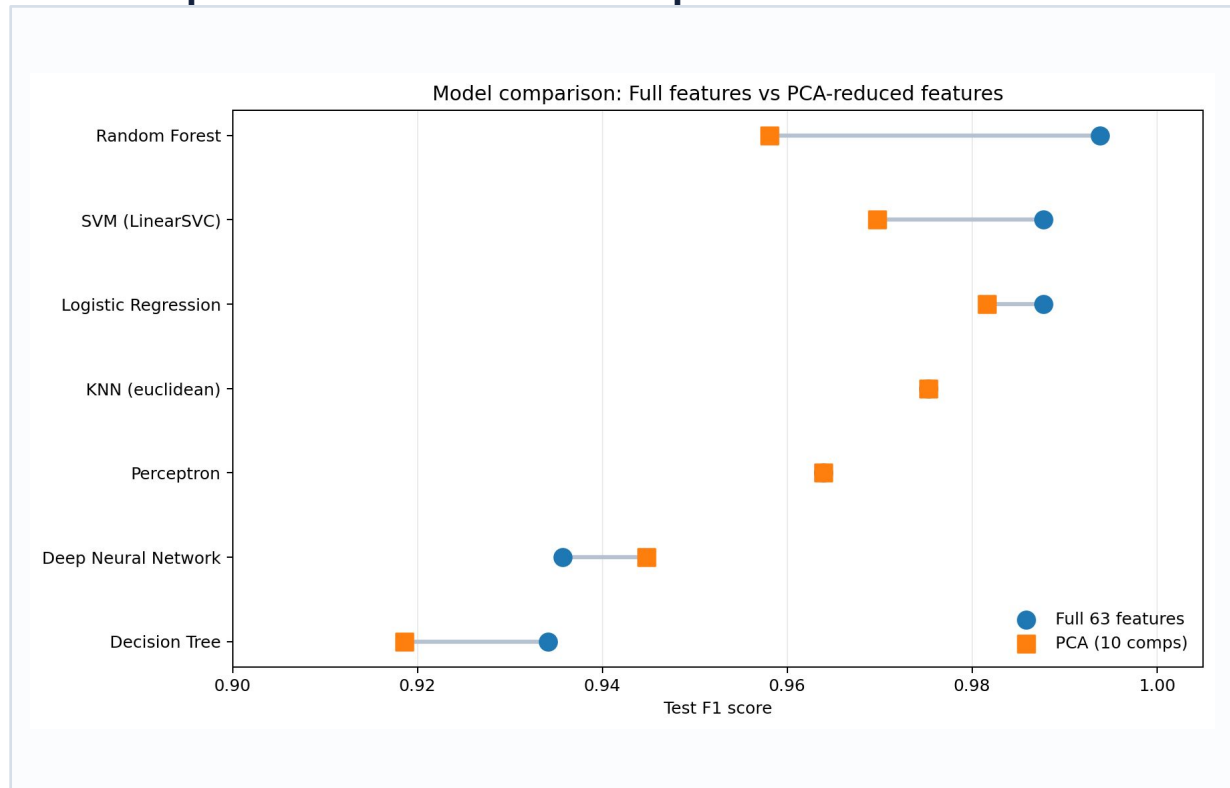
Data split



Why F1 mattered

- Accuracy gives an easy overall summary.
- Precision and recall show different error trade-offs.
- F1 score let us compare models with one balanced classification metric, so it became our main ranking number in the results slide.

Supervised learning clearly outperformed the earlier unsupervised attempt, and the best model depended on whether we kept all features or used PCA.

**0.9938**Random Forest test F1
(full 63 features)**0.9816**Logistic Regression test
F1
(PCA, 10 comps)**Key finding**

Random Forest was the strongest raw baseline, while Logistic Regression became the best reduced-feature model and the cleaner final choice.

The chart compares test F1 on the full feature set vs. the PCA-reduced feature set.

```
=== Baseline Model Results (sorted by Test F1) ===
```

Model	Train Acc	Train Prec	Train Rec	Train F1	Test Acc	Test Prec	Test Rec	Test F1
Random Forest	1.0000	1.0000	1.0000	1.0000	0.9945	1.0000	0.9877	0.9938
Logistic Regression	0.9976	1.0000	0.9947	0.9974	0.9890	0.9877	0.9877	0.9877
SVM (LinearSVC)	1.0000	1.0000	1.0000	1.0000	0.9890	0.9877	0.9877	0.9877
KNN (euclidean)	0.9953	0.9947	0.9947	0.9947	0.9779	0.9753	0.9753	0.9753
Perceptron	0.9929	0.9895	0.9947	0.9921	0.9669	0.9412	0.9877	0.9639
Deep Neural Network	0.9929	0.9845	1.0000	0.9922	0.9392	0.8889	0.9877	0.9357
Decision Tree	1.0000	1.0000	1.0000	1.0000	0.9392	0.9070	0.9630	0.9341

```
=== PCA Reduced Results (sorted by Test F1) ===
```

Model	Train Acc	Train Prec	Train Rec	Train F1	Test Acc	Test Prec	Test Rec	Test F1
Logistic Regression	0.9976	1.0000	0.9947	0.9974	0.9834	0.9756	0.9877	0.9816
KNN (euclidean)	0.9953	0.9947	0.9947	0.9947	0.9779	0.9753	0.9753	0.9753
SVM (LinearSVC)	1.0000	1.0000	1.0000	1.0000	0.9724	0.9524	0.9877	0.9697
Perceptron	0.9953	0.9947	0.9947	0.9947	0.9669	0.9412	0.9877	0.9639
Random Forest	1.0000	1.0000	1.0000	1.0000	0.9613	0.9302	0.9877	0.9581
Deep Neural Network	0.9787	0.9892	0.9632	0.9760	0.9503	0.9390	0.9506	0.9448
Decision Tree	1.0000	1.0000	1.0000	1.0000	0.9227	0.8681	0.9753	0.9186

Main conclusion: the landmark coordinates carry enough signal that a simple supervised classifier can solve the left-vs-right task extremely well.

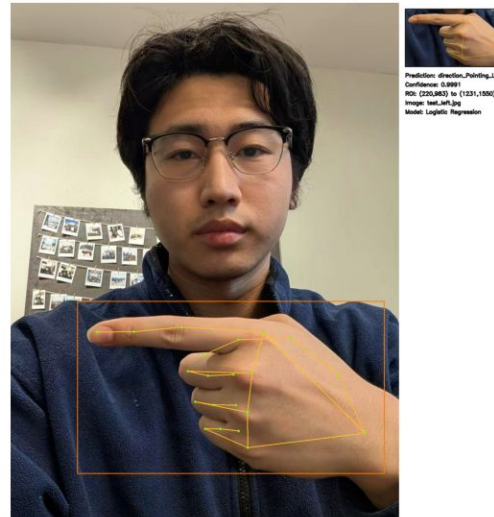
What we learned

- Unsupervised exploration (t-SNE + K-means) was useful for understanding the data, but not strong enough as the final predictive approach.
- Random Forest achieved the best full-feature benchmark performance.
- Logistic Regression performed the best after PCA and gave us a simple, accurate final story.

Future work

- Collect more gestures and more users for better generalization.
- Try temporal models on video sequences instead of single poses.
- Improve robustness to camera angle, lighting, and partial occlusion.

Hand landmarks + ROI + model prediction



Prediction: direction_Pointing_Left
Confidence: 0.9991
ROI: (220,603) to (331,1080)
Image: test_left.jpg
Model: Logistic Regression

Predicted class: direction_Pointing_Left
Confidence : 0.9991
Model : Logistic Regression

Hand landmarks + ROI + model prediction



Prediction: direction_Pointing_Right
Confidence: 0.7398
ROI: (572,1016) to (1140,1428)
Image: test_right.jpg
Model: Logistic Regression
Hand: Unknown

Predicted class: direction_Pointing_Right
Confidence : 0.7398
Model : Logistic Regression